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# Linear Algebra (Sketch)

"Vector": an 'N-tuple' (list) of numbers.

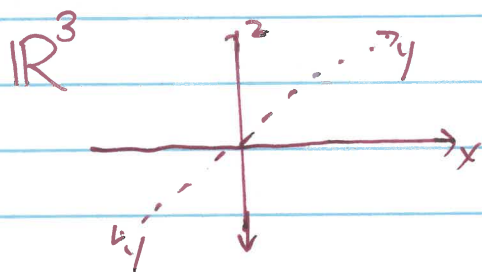
Examples:  $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$   $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$   $(5)$

$2 \times 1$   $3 \times 1$   $1 \times 1$

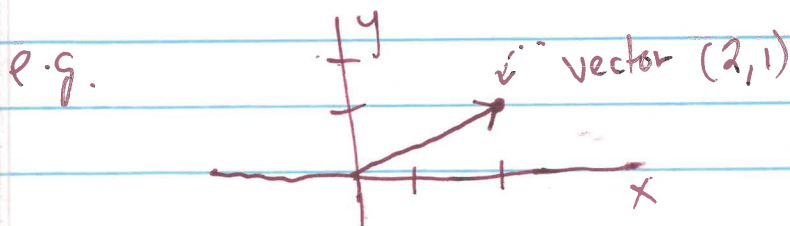
$\uparrow$   
a scalar  
as special case

" $\mathbb{R}^N$ ": The set of all N-tuples (vectors) of real numbers.

$\mathbb{R}^2$ : the plane



Vectors can be viewed as points in  $\mathbb{R}^N$



# Linear Algebra

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## Vector operations

Addition : Add vector components element-wise

Example  $\begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$

Multiplication by scalar : Multiply all elements by scalar.

Example:  $2 \cdot \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \\ 2 \end{pmatrix}$

Subtraction : Add -1 times vector (or subtract element-wise!)

Example :  $\begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$

$$= \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} + \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}$$

Zero identity :  $\begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}$

Anything + zero vector  
gives original vector back

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Dot product & length (or "norm")

Let  $a$  and  $b$  be vectors in  $\mathbb{R}^N$

'Dot product'  $a \cdot b = \sum_{n=1}^N a_n b_n$  a scalar

Example  $\begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} = a \quad \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = b$

$a \cdot b = \overset{12}{4 \times 3} + \overset{0}{-2 \times 0} + \overset{-1}{1 \times -1} = 11$

" $L_2$  norm" = length of vector (Euclidean)  
in  $\mathbb{R}^N$

$$\|a\|_2 = \sqrt{a_1^2 + a_2^2 + \dots + a_N^2}$$

Note that  $a \cdot a = \sum_{n=1}^N a_n^2$

so  $\|a\|_2 = \sqrt{a \cdot a}$

" $L_1$  norm" =  $\|a\|_1 = \sum_{n=1}^N |a_n|$  "Manhattan Distance"

Distributive property

Let  $x, y$  be vectors in  $\mathbb{R}^N$  and  $c$  a scalar  
then  $c(x+y) = cx + cy$ .

★  
Important  
in  
machine  
learning.

# Linear Algebra

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## Dot product and angle between vectors

Let  $v, w$  be vectors in  $\mathbb{R}^N$  and  $\theta$  be angle between them.

$$\cos \theta = \frac{v \cdot w}{\|v\| \|w\|} \quad \text{or} \quad v \cdot w = \|v\| \|w\| \cos \theta$$

Example:  $v = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$   $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

What is ~~cos~~  $\theta$ ? 1) Use intuition

2) confirm w/ result.

$$\theta = 90^\circ$$

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"Span" of set of vectors

Let  $v, w$  be vectors in  $\mathbb{R}^N$

The span of  $v, w$  equals the set of all vectors  $x$ ,  $x = av + bw$ , where  $a, b$  are scalars. All linear combinations of  $v, w$ .

Example: What is span of  $\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ ?

$$\text{All } x = a \begin{pmatrix} 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

This is the entire  $x, y$  plane!

How about  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -3 \\ 3 \end{pmatrix}$ ?

Same.

How about  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ ?

In this case  $\begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$  so

$\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}$  point in same direction

$\Rightarrow$  Span is line through  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$  and  $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$

$\hookrightarrow$  only one dimension!

We say  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}$  are linearly dependent.

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Let  $x_1, x_2, \dots, x_N$  be vectors in  $\mathbb{R}^N$

Then  $\text{Span}(x_1, x_2, \dots, x_N) = \mathbb{R}^N$

if and only if  $x_1, x_2, \dots, x_N$  are

linearly independent (no  $x_m$  can be obtained as  
linear combination of other  $x_1, \dots, x_{m-1},$   
 $x_{m+1}, \dots, x_N$ )

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Row vector / Column vector

$(3, 2, 1)$

("1 x 3")

$\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

("3 x 1")

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Matrix

Matrix = An  $m \times n$  'array' of  
real numbers

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$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \leftarrow \begin{matrix} a \\ 3 \times 3 \\ \text{matrix} \end{matrix}$$

$a_{ij}$  =  $i, j$ th  
"element" of  $A$

Example:  $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$I_{22} = 1$

$I_{21} = 0$

3x3 "identity matrix"

Example

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_2^2 \end{pmatrix}$$

2x2 "Covariance  
Matrix"

Note  $\sigma_{2,1} = \sigma_{1,2}$

Matrices do not have to be "square"

Example

$$B = \begin{pmatrix} 3 & -1 & 0 \\ 4 & 2 & 4 \end{pmatrix}$$

"2x3 matrix"

Matrices: Row vectors vs Column vectors

Same matrix as bottom of last page

$$B = \begin{pmatrix} 3 & -1 & 0 \\ 4 & 2 & 4 \end{pmatrix}$$

→ in  $\mathbb{R}^3$   
=

B has: 2 "row vectors"

3 "column vectors"

→ in  $\mathbb{R}^2$   
=

Question: What is span of the 3 column vectors? ~~is it~~

Matrix Addition / Subtraction

Let  $A, B$  both be  $m \times n$  matrices

→ Must have same dimensions!

$$C = A + B = \begin{bmatrix} a_{11} + b_{11} & a_{1n} + b_{1n} \\ a_{21} + b_{21} & \vdots \\ \vdots & \vdots \\ a_{m1} + b_{m1} & a_{mn} + b_{mn} \end{bmatrix}$$

Just add / subtract element-wise.

Example:

$$\begin{pmatrix} 3 & 7 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} -2 & 1 \\ -8 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 8 \\ -9 & 4 \end{pmatrix}$$

Matrix multiplication by scalar:

$$cA = \begin{bmatrix} ca_{11} & ca_{1n} \\ ca_{m1} & ca_{mn} \end{bmatrix} \quad \text{multiply all elements by } c$$

e.g.

$$2 \begin{pmatrix} 3 & 7 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 6 & 14 \\ -2 & 0 \end{pmatrix}$$

Matrix transpose

Reverse row and column indices.

$$A^T \text{ or } A' = [A_{ji}]$$

Alt. notation

e.g.

$$\begin{bmatrix} 3 & 7 \\ -1 & 0 \end{bmatrix}' = \begin{bmatrix} 3 & -1 \\ 7 & 0 \end{bmatrix}$$

Symmetric Matrix  $\Rightarrow$  A matrix that is equal to its transpose. Must be "square" ( $N \times N$ )

Example:  $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$I_3' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \text{symmetric.}$

Example

$\begin{pmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_2^2 \end{pmatrix}$

covariance matrix

(b/c  $\sigma_{1,2} = \sigma_{2,1}$ )

( $\text{cov}(X,Y) = \text{cov}(Y,X)$ )

## \* Matrix multiplication

$A \quad B = C$   
 $m \times n \quad n \times p \quad m \times p$

$\uparrow \quad \uparrow$   
 must be equal!

where  $C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$

dot product of  
 $i^{\text{th}}$  row of A  
 and  $j^{\text{th}}$  column  
 of B

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## Example

$$\begin{pmatrix} 2 & -3 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{bmatrix} -3 & -1 \\ 0 & 1 \end{bmatrix}$$

A                      B                      C

$$c_{11} = 2(0) + -3(1) = -3$$

$$c_{21} = 1(0) + 0(1) = 0$$

$$c_{12} = 2(1) + -3(1) = -1$$

$$c_{22} = 1(1) + 0(1) = 1$$

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## Matrix inverse

Motivation solve for  $x$ :  $2x = 4$

$$\frac{1}{2} \cdot 2x = \frac{1}{2} \cdot 4$$

$\frac{1}{2}$   
inverse!

Matrix equation:

$$x = 2$$

$$\begin{matrix} A & x & = & C \\ n \times n & n \times 1 & & n \times 1 \end{matrix}$$

System of  $n$  equations

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If there exists an  $n \times n$  matrix  $B$  such that:  $BA = I = AB$  then we call  $B$  the inverse of  $A$  and denote it " $A^{-1}$ ".

## Example

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 10 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 12 \\ 12 \\ 10 \end{pmatrix}$$

$A \quad \quad \quad X \quad \quad \quad C$

check it!

$$A^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{10} \end{pmatrix} \quad \Rightarrow \quad X = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{10} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 10 \end{pmatrix}$$

$$\Rightarrow \underbrace{A^{-1}A} X = A^{-1}C$$

$$\underbrace{I} \Rightarrow X = A^{-1}C$$

$$X = \begin{pmatrix} 3 \\ 6 \\ 1 \end{pmatrix}$$

## Quadratic Form

Let  $\Sigma$  be  $N \times N$  symmetric matrix

(Think Covariance matrix)

$W = N \times 1$  vector

$$\text{Then } \underset{(1 \times N)}{W'} \underset{(N \times N)}{\Sigma} \underset{(N \times 1)}{W} = \underset{(1 \times 1)}{\text{a scalar!}}$$

and  $W' \Sigma W \geq 0$  and  $= 0$  only if

"positive definite"  $W = 0$ .

Example

$$(w_1 \ w_2) \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

$\sigma_{21} \leftarrow \sigma_{12}$

$$= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{1,2}$$

Extends to  $N$  dimensions.   
 portfolio variance!